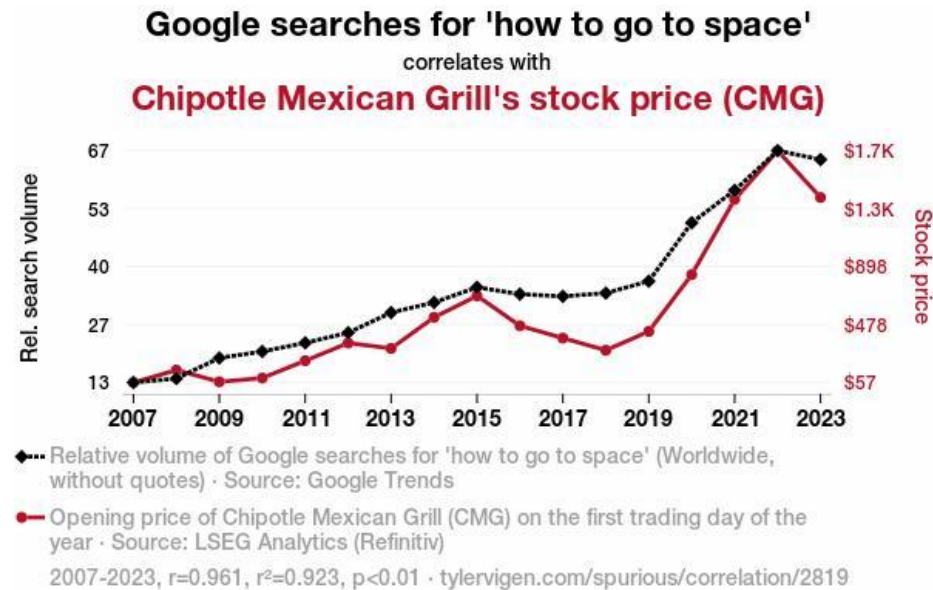


Relationships between two quantitative variables



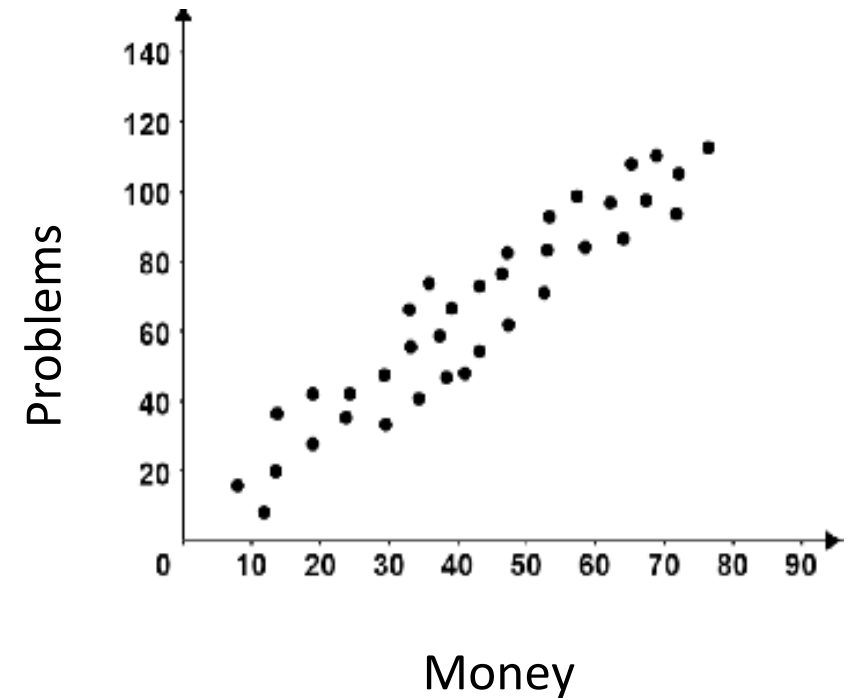
Overview

Review

- Quantiles and boxplots
- Scatter plots and correlation

Simple linear regression

If there is time: sampling and bias



Announcement: homework 2

Homework 3 has been posted!

It is due on Gradescope on **Sunday
September 27th at 11pm**

- **Be sure to mark each question on
Gradescope!**



Keep attending the practice sessions!

Announcement: mini-exam

The mini-exam will **in class** on **Tuesday September 29th**

- Last 30 minutes of class

You should know/understand:

1. All the symbols we have used in class represent
 - E.g., A professor believes the average height of Yale students is 69 inches. How can you write this using the symbols we discussed in class?
2. Should be able to answer simple questions about the R code we have used
 - E.g., What does this R code do? `boxplot(my_vec)`

	Statistic	Parameter
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Proportion	$\hat{p}$	π
Correlation	r	ρ
regression slope	b	β

If you have accommodations for extra time, you will take the exam in Kline tower, 11th floor , room 1105B

Review

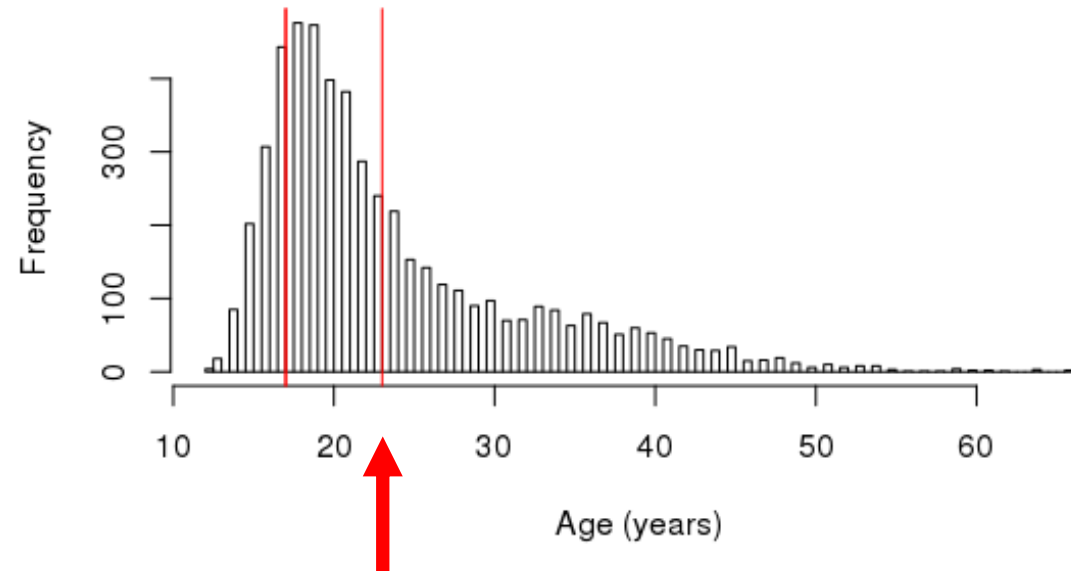
Review: quantiles (percentiles)

The **p^{th} percentile** is a value **x** which is greater than p percent of the data

- i.e., $p\%$ of the data is less than x

Let's look at the age people were arrested for using marijuana in Toronto

Histogram of Ages of people arrested for marijuana use



60th percentile value is 23

i.e., 60% of the arrested were of ages 23 or less

In R: `quantile(Arrests$age, .6)`

Preview: Quantiles of the normal distribution

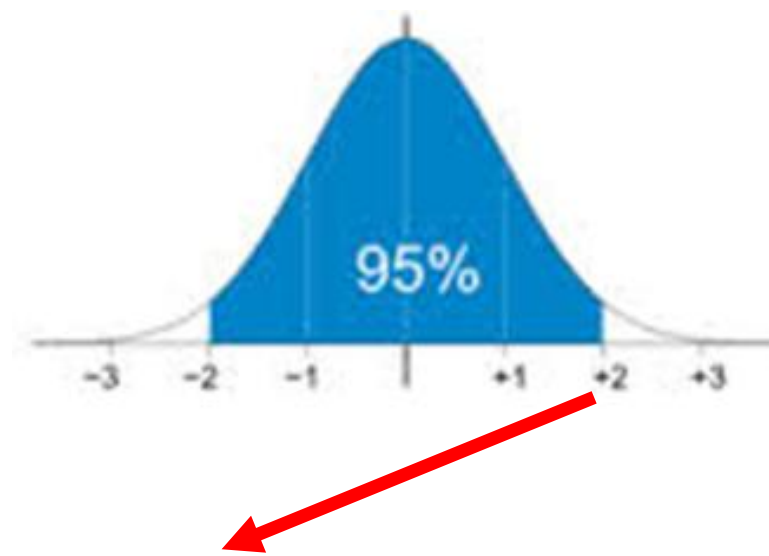
We can also get quantiles of a normal distribution

Question: Suppose our data has a (standard) normal distribution. What is the 97.5% percentile of data?

- i.e., What z-score value is greater than 97.5% of the data in a standard normal distribution?

Answer: 2!

- (Or really 1.96 to be more precise)



97.5% of values are less than 2

In R: `qnorm(.975)`

The quantile universe

Five-Number Summary = (minimum, Q_1 , median, Q_3 , maximum)

Q_1 = 25th percentile, Q_3 = 75th percentile

Range = maximum – minimum

Interquartile range (IQR) = $Q_3 - Q_1$

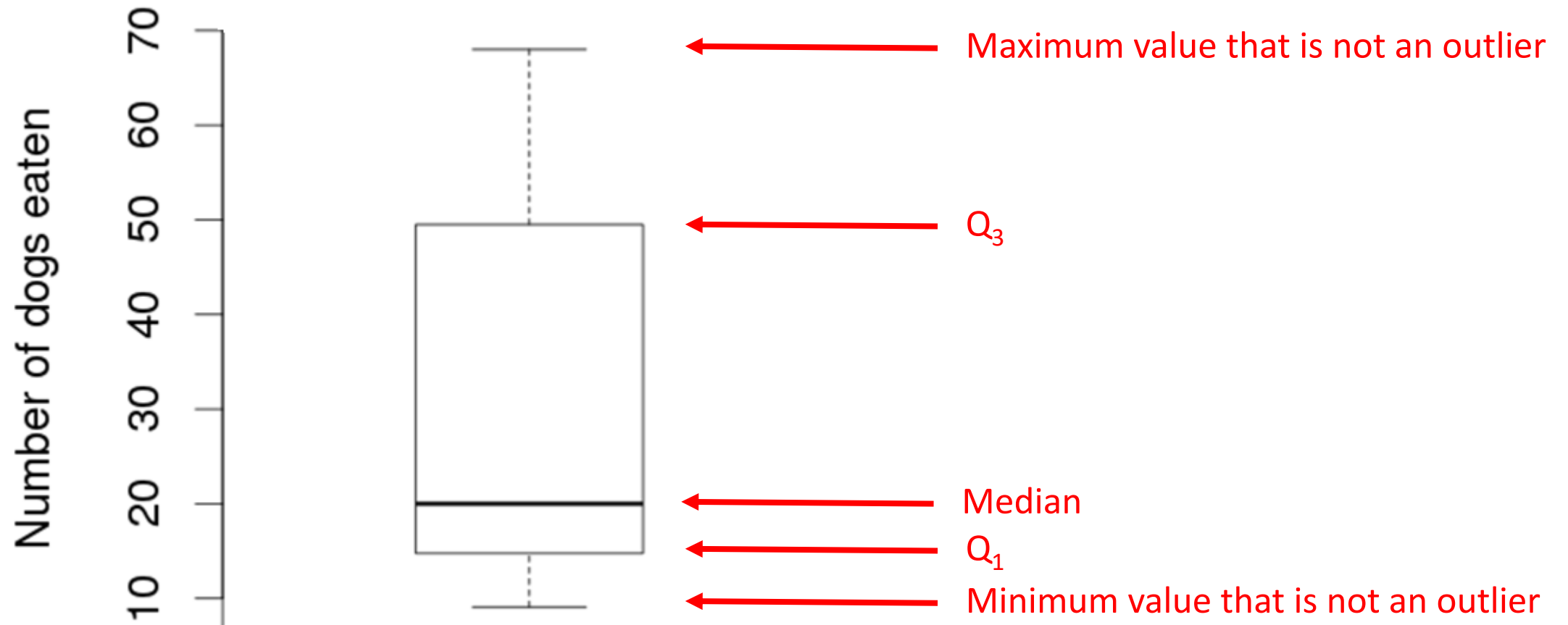
As a rule of thumb, we call a data value an **outlier** if it is:

Smaller than: $Q_1 - 1.5 * IQR$

Larger than: $Q_3 + 1.5 * IQR$

In R: `fivenum(v)`

Box plot of the number of hot dogs eaten by the men's contest winners 1980 to 2010

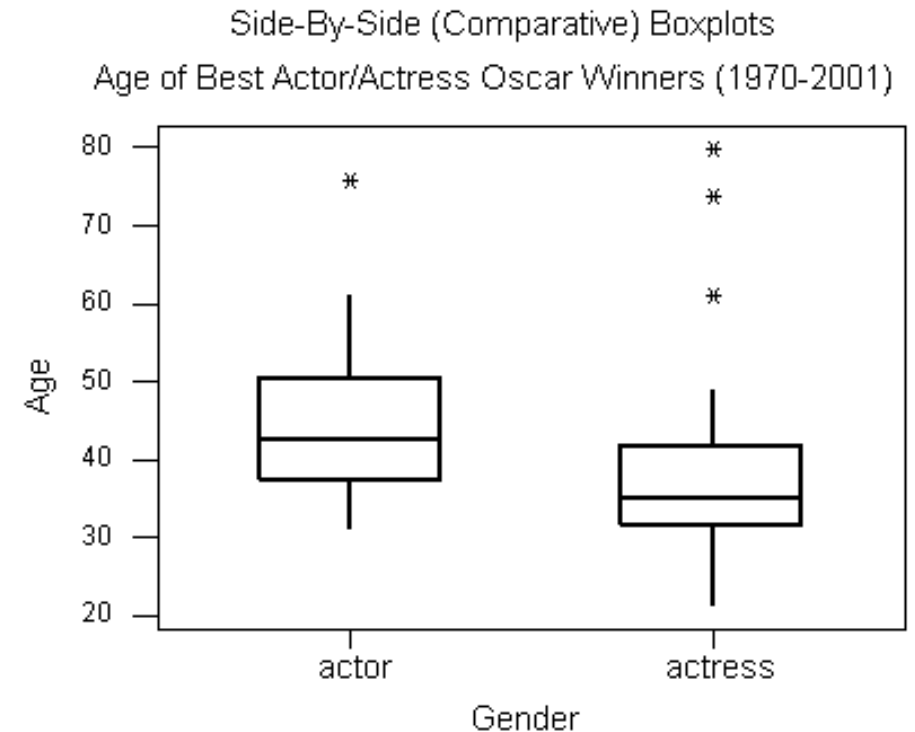


R: `boxplot(v)`

Comparing quantitative variables across categories

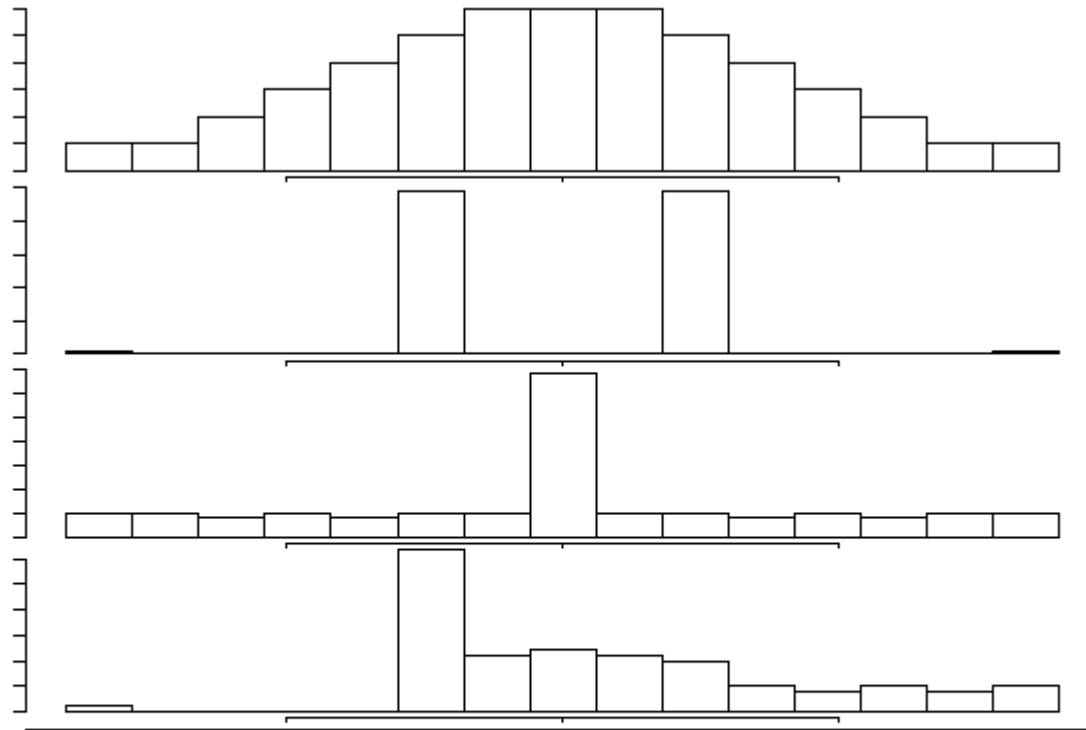
Often one wants to compare quantitative variables across categories

Side-by-Side graphs are a way to visually compare quantitative variables across different categories



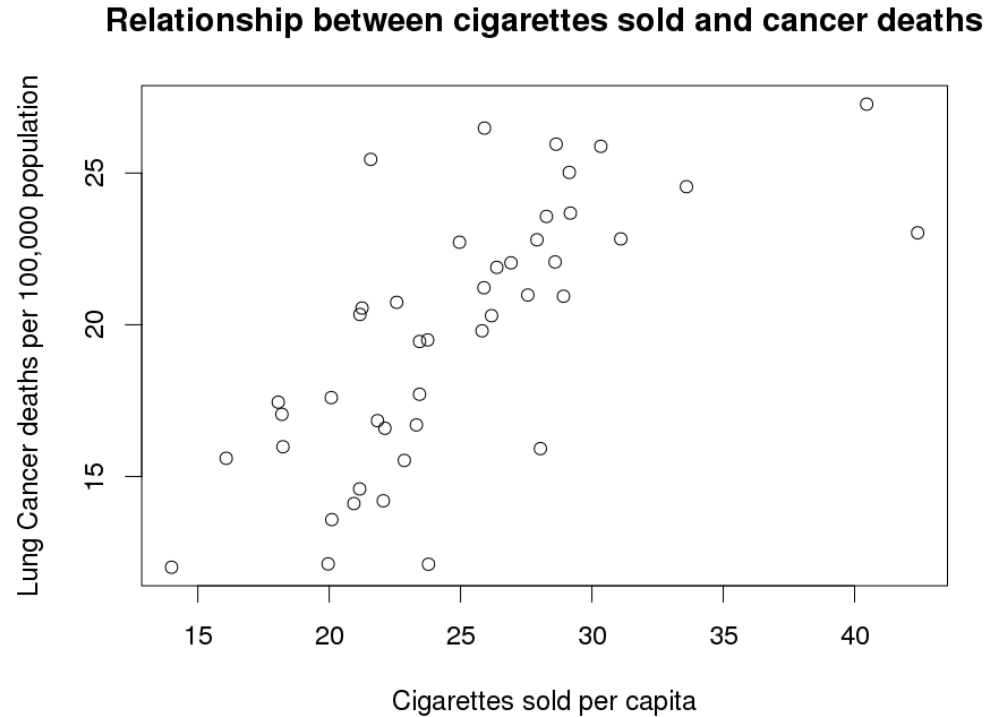
```
boxplot(v1, v2, names = c("name 1", "name 2"), ylab = "y-axis name")
```

Box plots don't capture everything



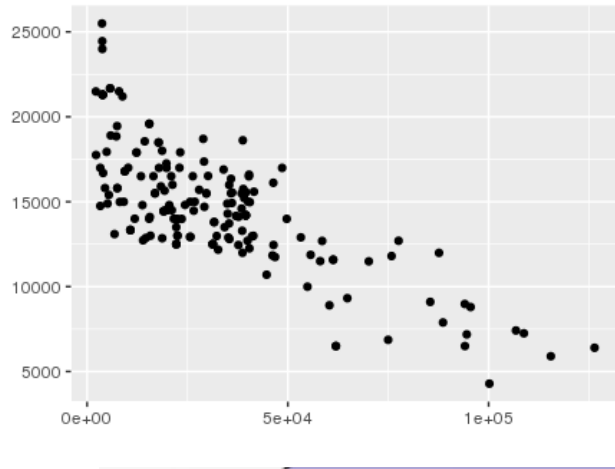
Do you think the box plots for these distributions look similar?

Review: scatter plots and the correlation coefficient

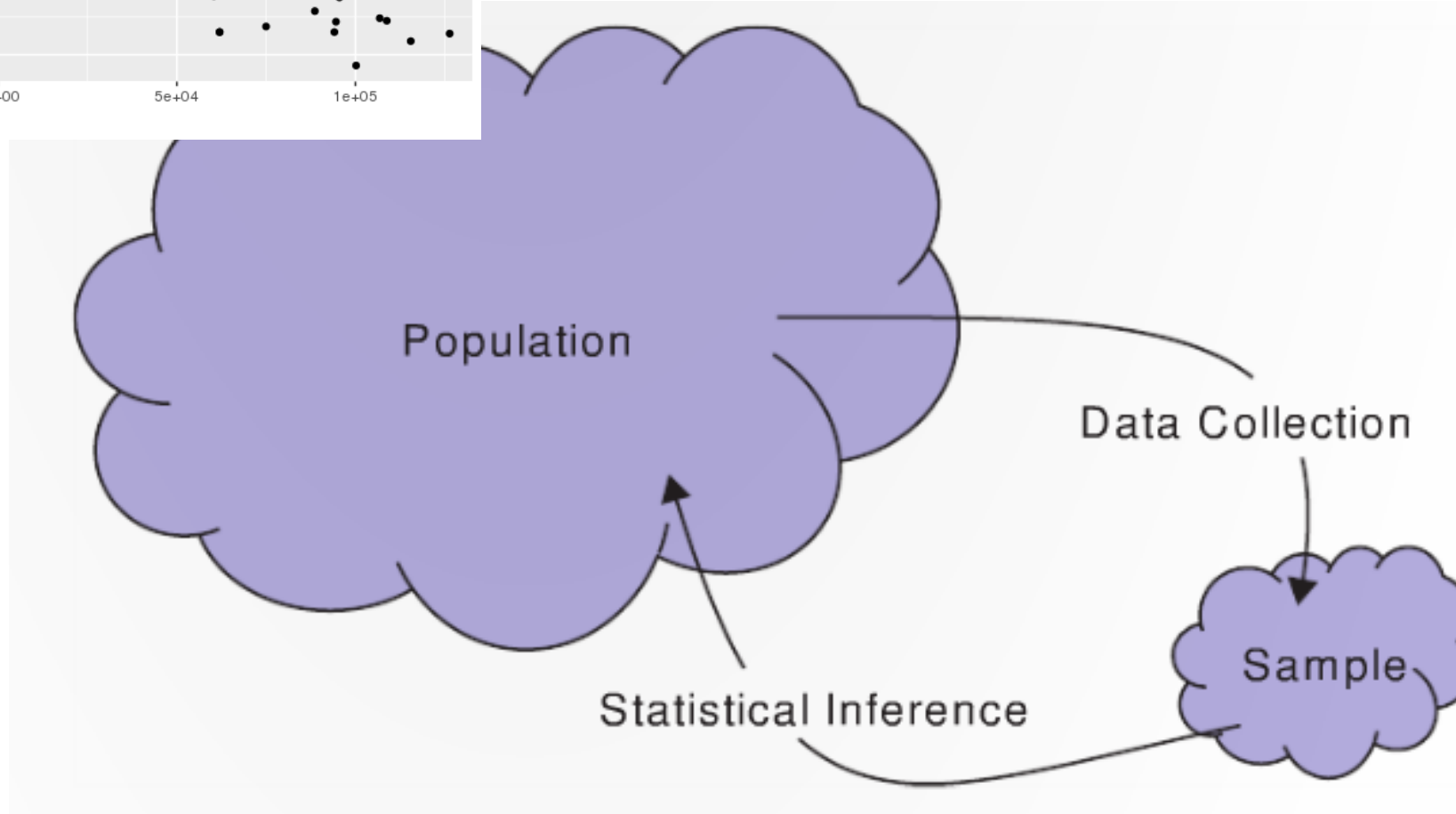


$$r = \frac{1}{(n - 1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

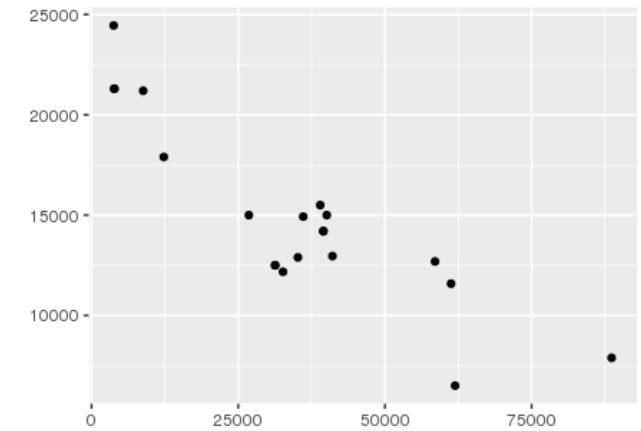
The **correlation** is measure of the strength and direction of a linear association between two variables



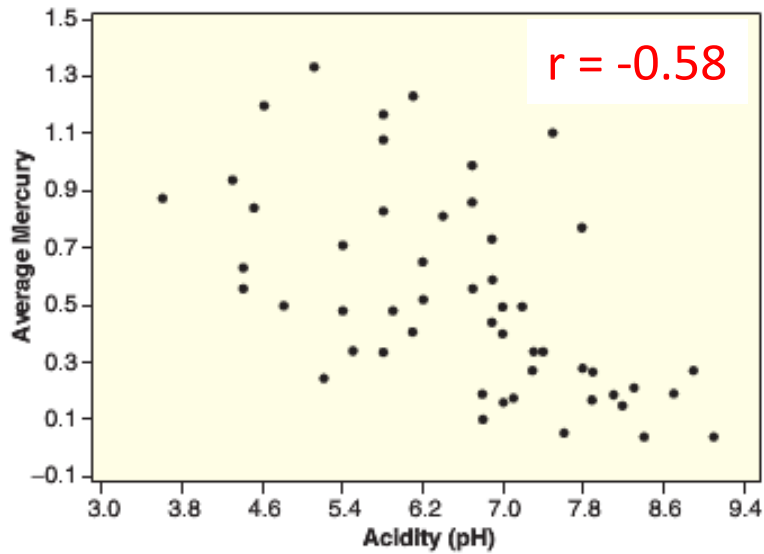
ρ parameter



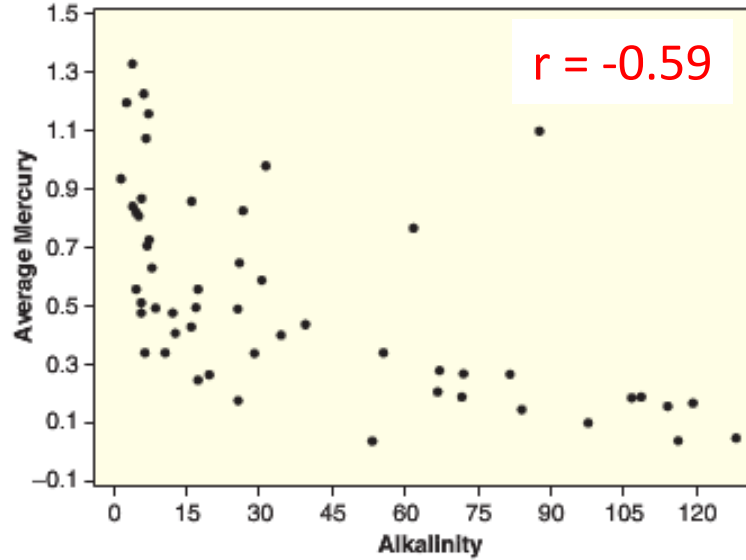
r statistic



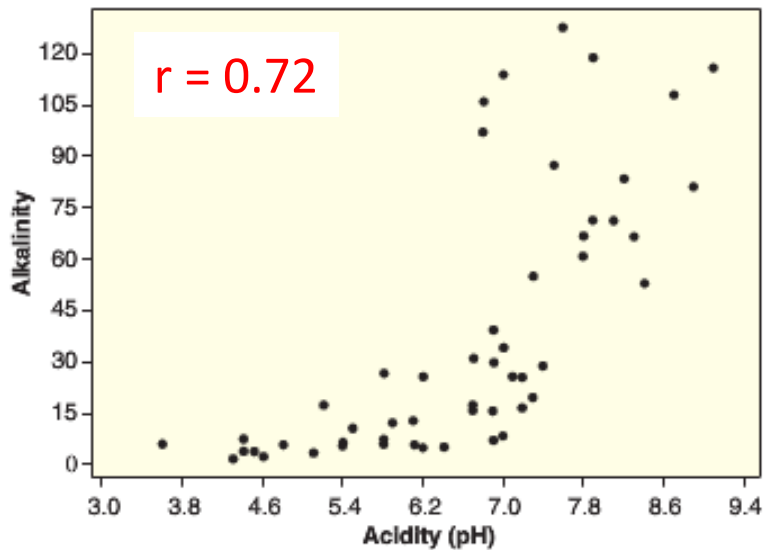
Florida lakes



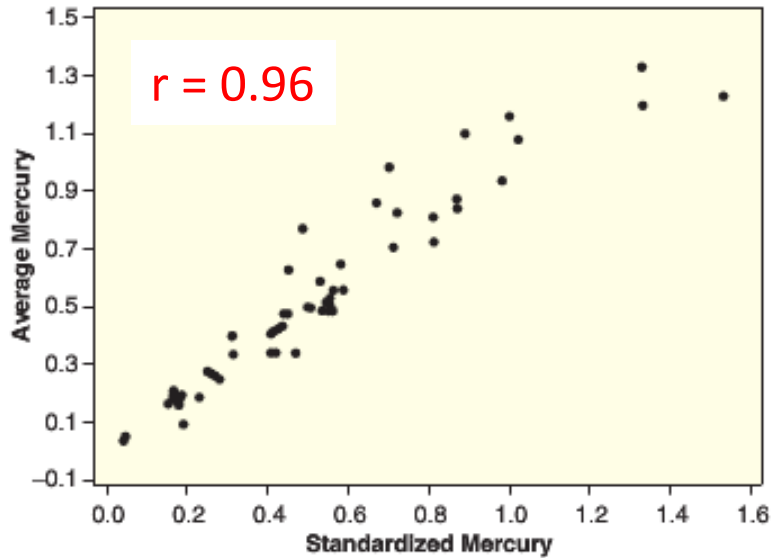
(a) Average mercury level vs acidity



(b) Average mercury level vs alkalinity



(c) Alkalinity vs acidity



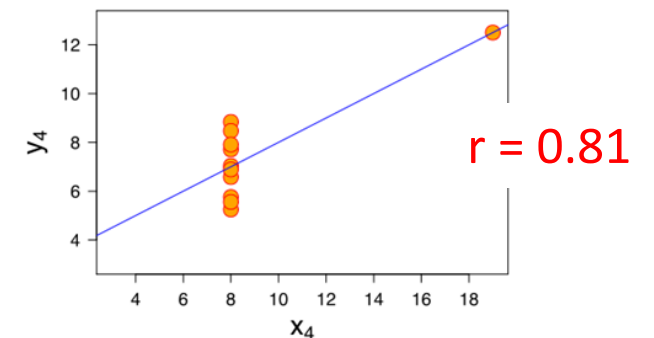
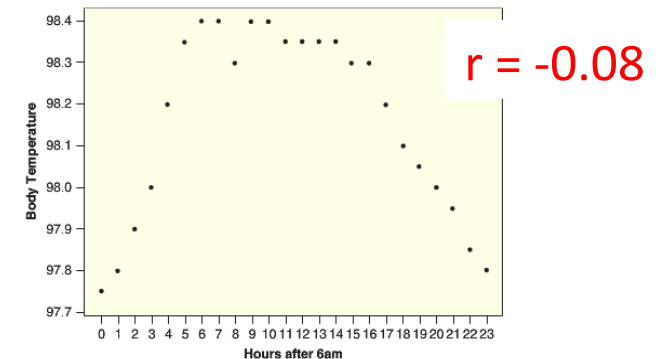
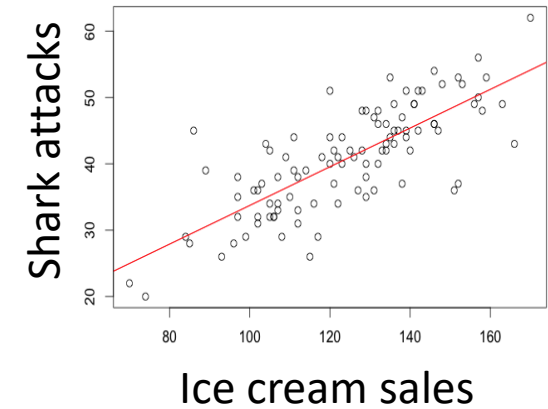
(d) Average vs standardized mercury levels

create a scatter plot
`plot(x, y)`

calculate the correlation
`cor(x, y)`

Correlation cautions

1. A strong positive or negative correlation does not (necessarily) imply a cause and effect relationship between two variables
2. A correlation near zero does not (necessarily) mean that two variables are not associated. Correlation only measures the strength of a linear relationship
3. Correlation can be heavily influenced by outliers. Always plot your data!



Linear regression

Regression

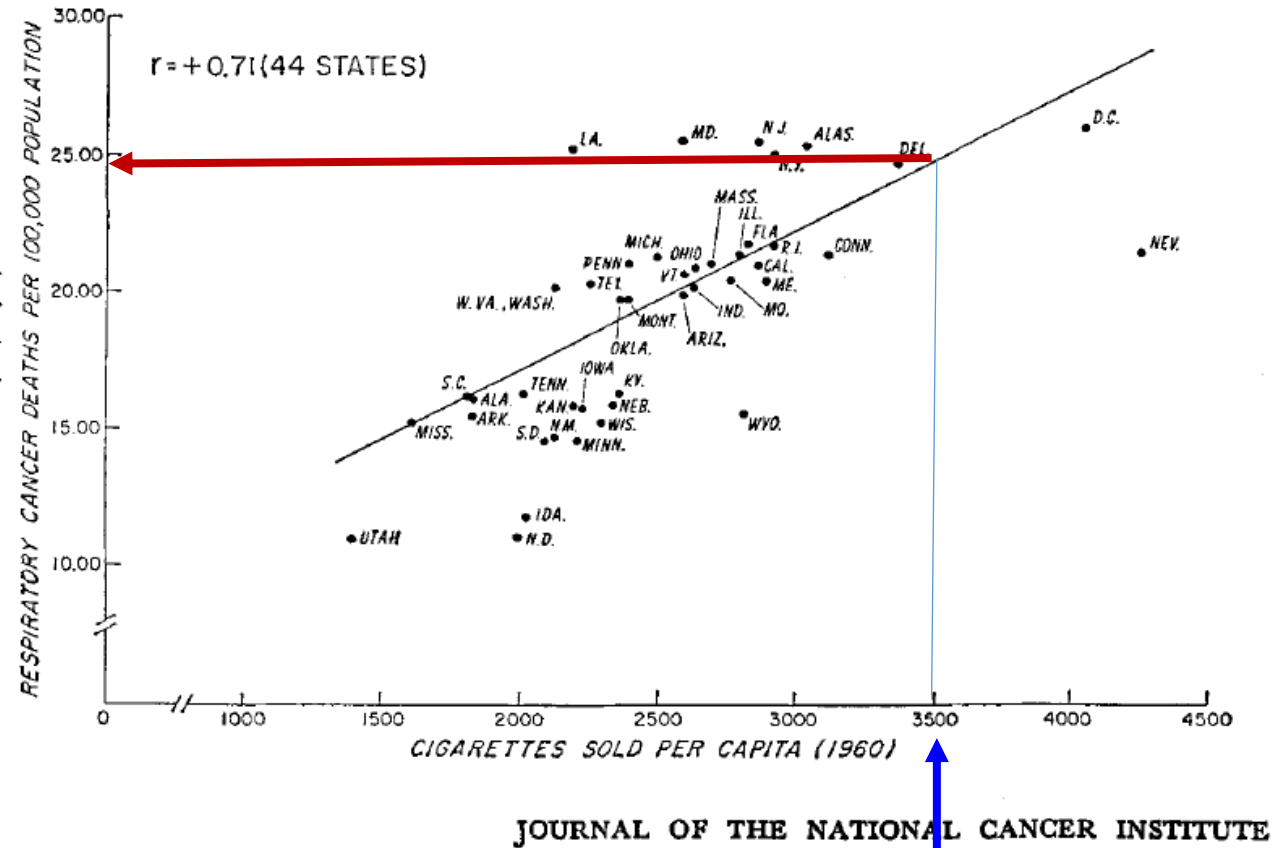
Regression is method of using one variable x to predict the value of a second variable y

- i.e., $\hat{y} = f(x)$

In **linear regression** we fit a line to the data, called the **regression line**

Cigarette cancer regression line

TEXT-FIGURE 2.—Correlation between average annual age-adjusted death rates for respiratory tract cancer (1956–61) and *per capita* cigarette sales (1960) in 44 States.

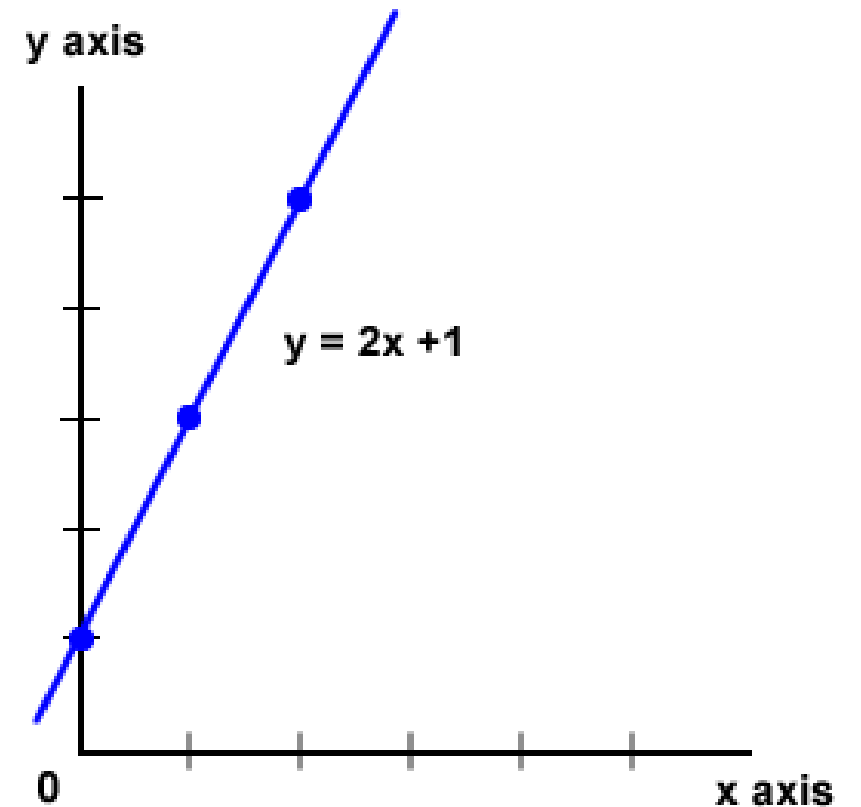


$$x_i = 3500$$

Equation for a line

What is the equation for a line?

$$\hat{y} = a + b \cdot x$$



Regression lines

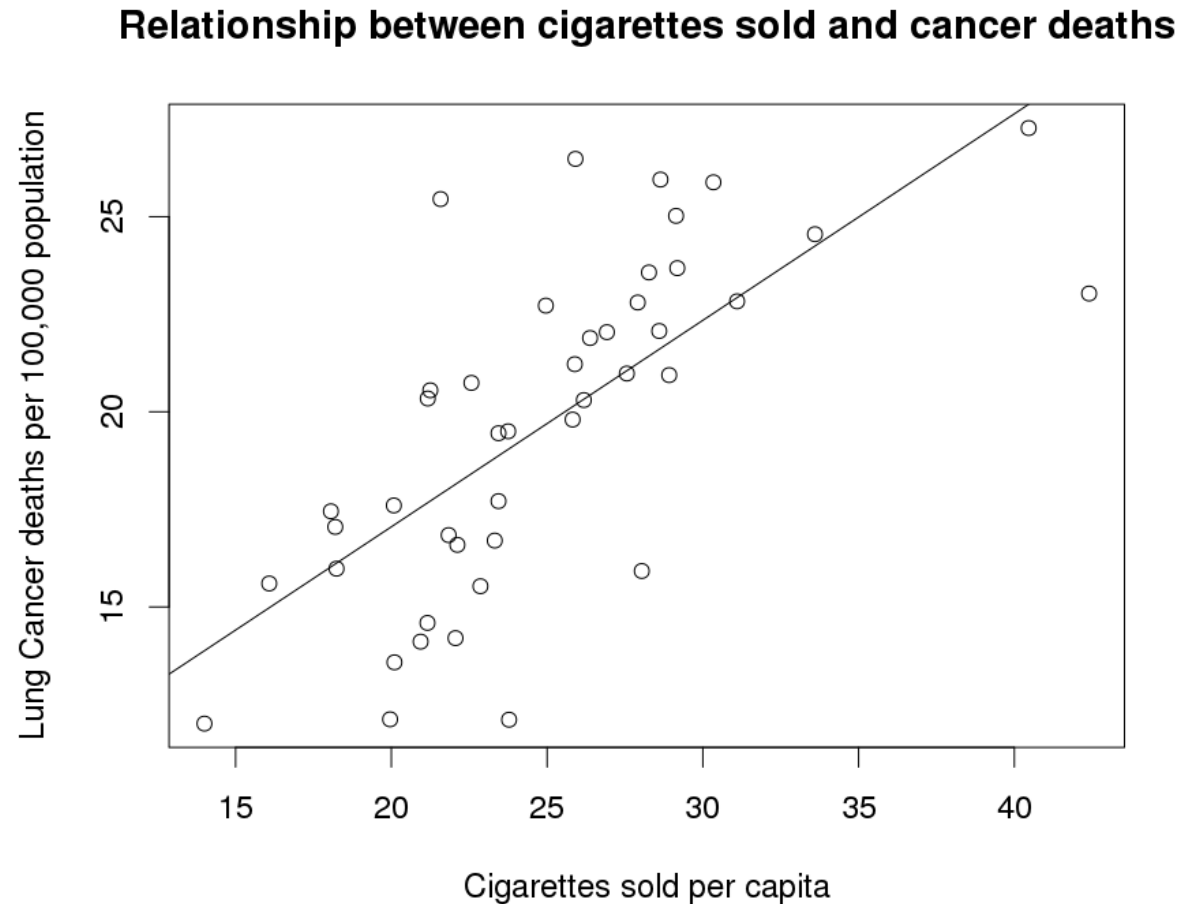
$$\hat{y} = a + b \cdot x$$

$$\textit{Response} = a + b \cdot \textit{Explanatory}$$

The slope ***b*** represents the predicted change in the response variable *y* given a one unit change in the explanatory variable *x*

The intercept ***a*** is the predicted value of the response variable *y* if the explanatory variable *x* were 0

Cancer smoking regression line



$$\hat{y} = a + b \cdot x$$

$$a = 6.47$$

$$b = 0.0053$$

$$R: \text{lm}(y \sim x)$$

$$\hat{y} = 6.47 + 0.0053 \cdot x$$

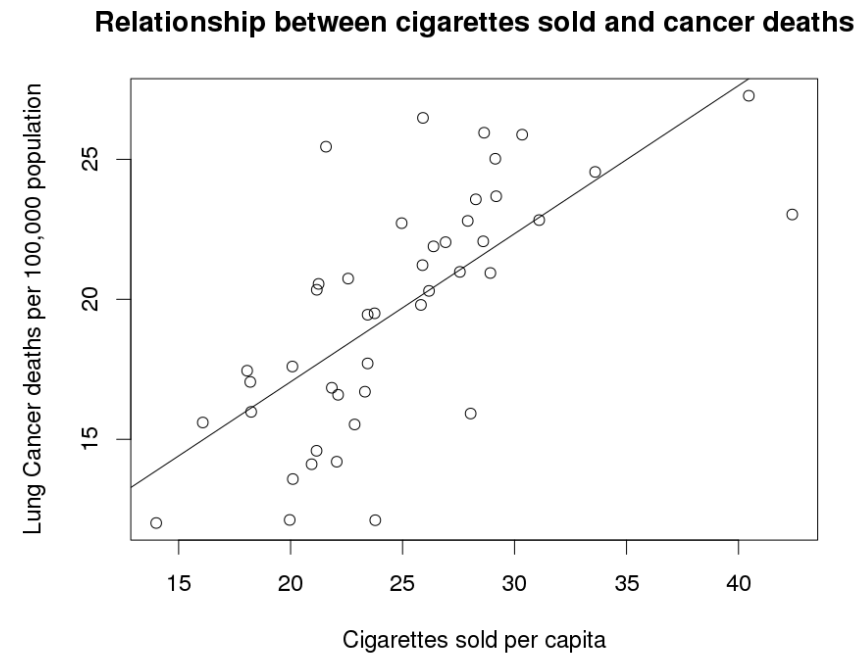
Using the regression line to make predictions

If a state sold 2,500 cigarettes per person

How many cancer deaths (per 100,000 people) would you expect?

$$a = 6.47, \quad b = .0053$$

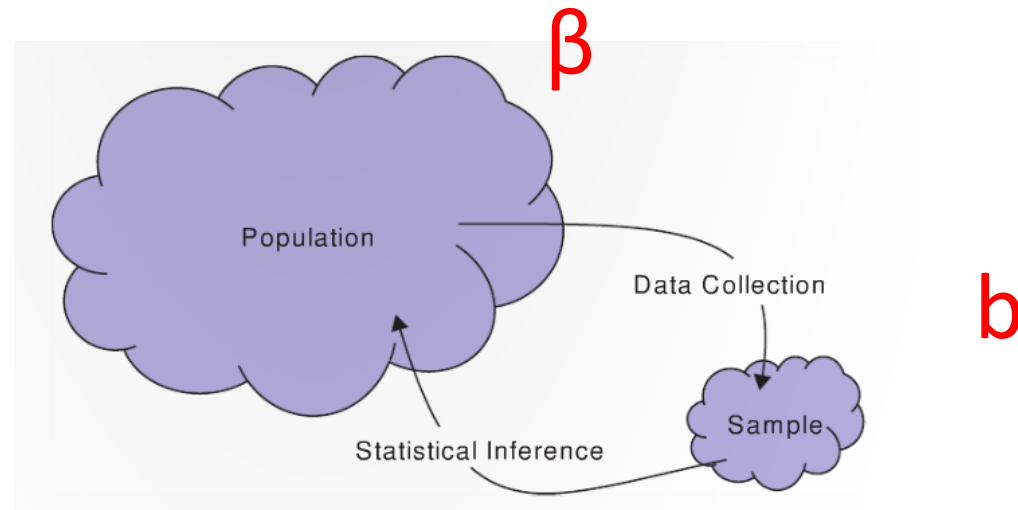
$$\hat{y} = 6.47 + .0053 \cdot x$$



Notation

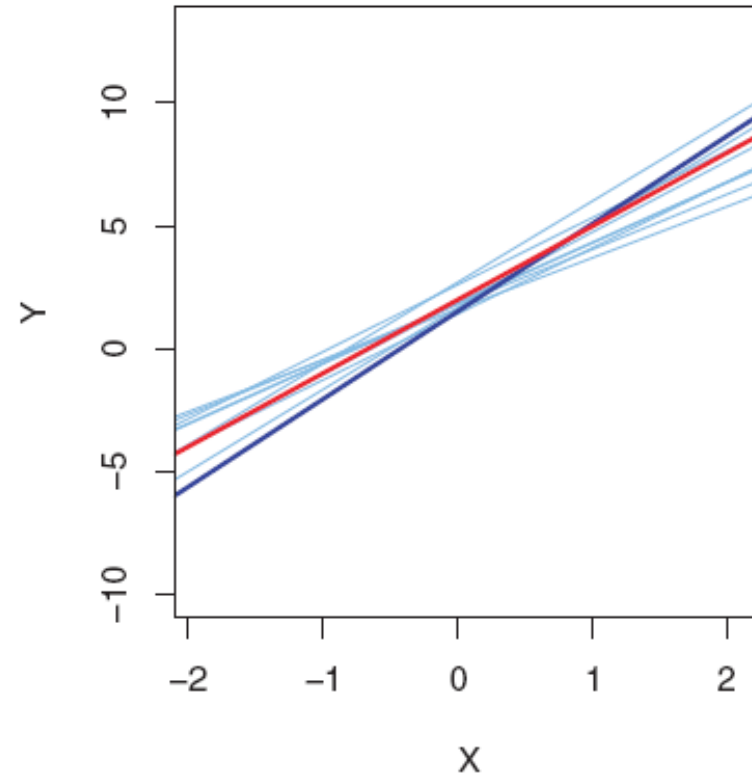
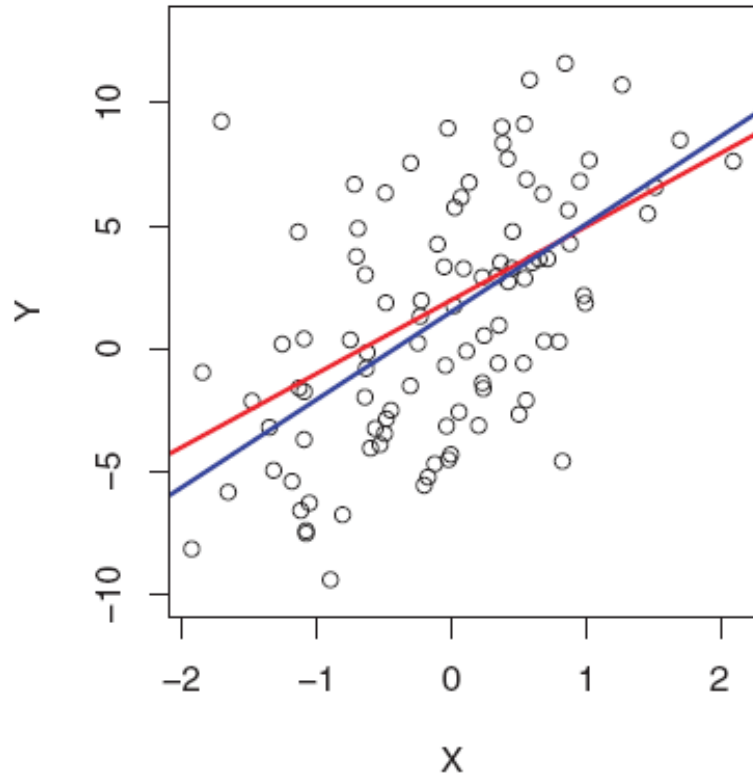
The letter **b** is typically used to denote the slope of the sample

The Greek letter **β** is used to denote the slope of the population



Population: β

Sample estimates: b



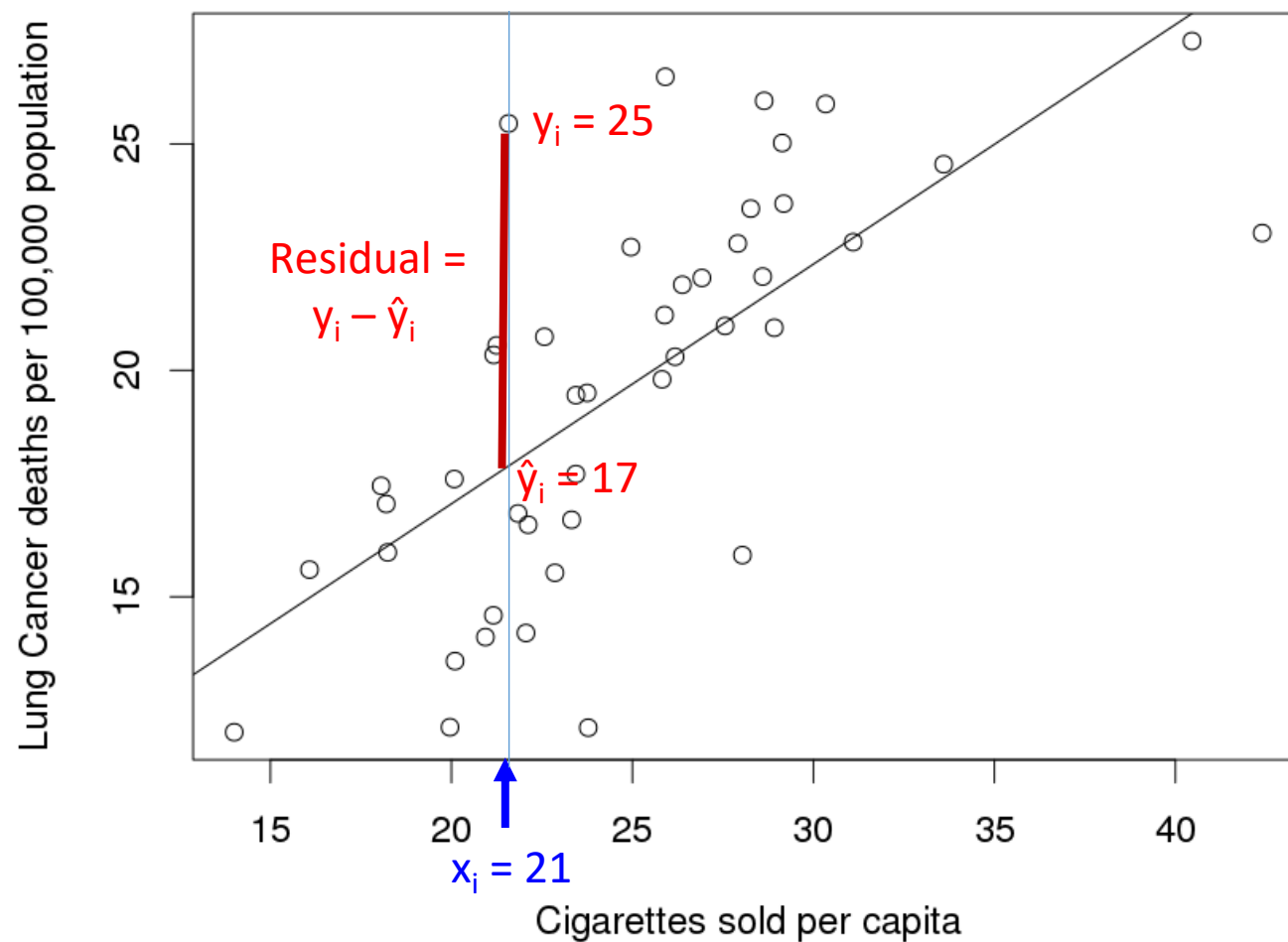
Residuals

The **residual** is the difference between an observed (y_i) and a predicted value ($\hat{y}_i$) of the response variable

$$Residual_i = Observed_i - Predicted_i = y_i - \hat{y}_i$$

Cancer smoking residuals

Relationship between cigarettes sold and cancer deaths



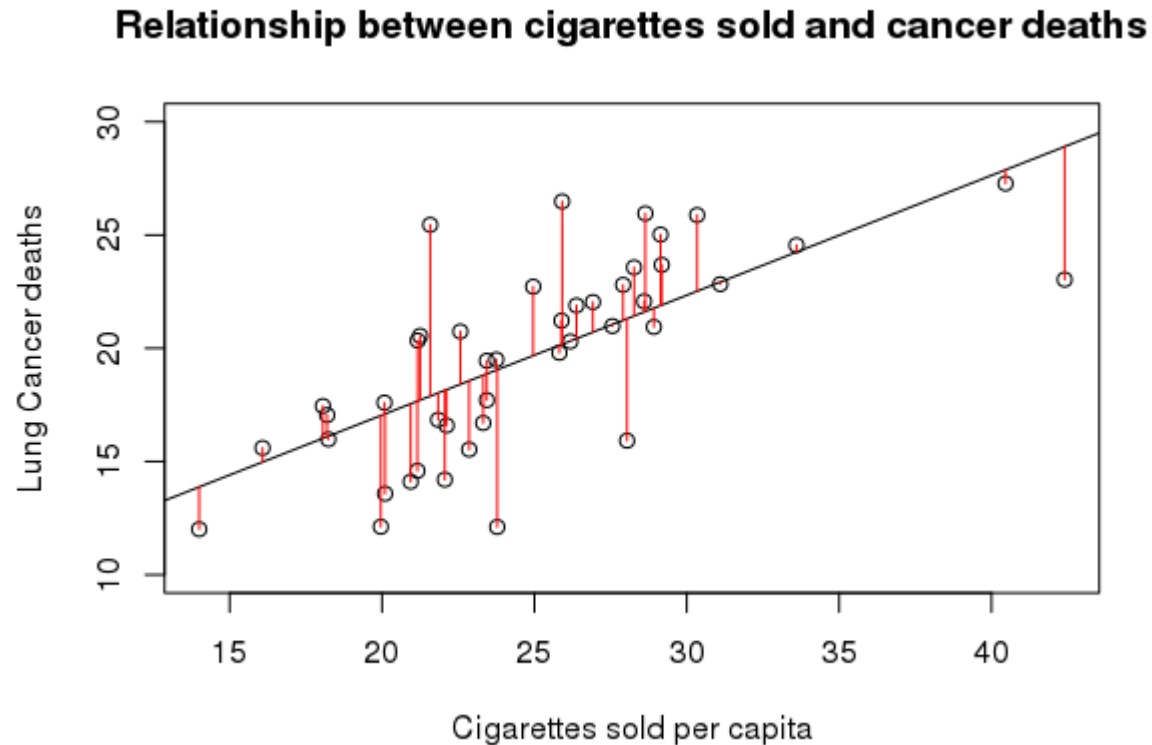
Cancer smoking residuals

$$\hat{y} = 6.47 + 0.0053 \cdot x$$


Cig per Capita (x)	Cancer obs (y)	Cancer pred ($\hat{y}$)	Residuals (y - $\hat{y}$)
1,820	17.05	16.10	0.95
2,582	19.80	20.13	-0.33
1,824	15.98	16.12	-0.14
2,860	22.07	21.60	0.47
3,110	22.83	22.93	-0.10
3,360	24.55	24.25	0.30
40,460	27.27	27.88	-0.61

Least squares line

The **least squares line**, also called '**the line of best fit**', is the line which minimizes the sum of squared residuals

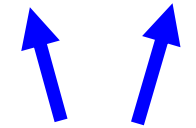


[Find the line of best fit](#)

Cancer smoking residuals

Cancer obs (y)	Cancer pred ($\hat{y}$)	Residuals (y - $\hat{y}$)	Residuals ² (y - $\hat{y}$) ²
17.05	16.10	0.95	0.90
19.80	20.13	-0.33	0.11
15.98	16.12	-0.14	0.02
22.07	21.60	0.47	0.22
22.83	22.93	-0.10	0.01
24.55	24.25	0.30	0.09
27.27	27.88	-0.61	0.37
23.57	21.24	2.14	4.59

$$\hat{y} = a + b \cdot x$$



Find the a and b



That minimizes the sum
of the squared residuals

Regression caution # 1

Avoid trying to apply the regression line to predict values far from those that were used to create the line

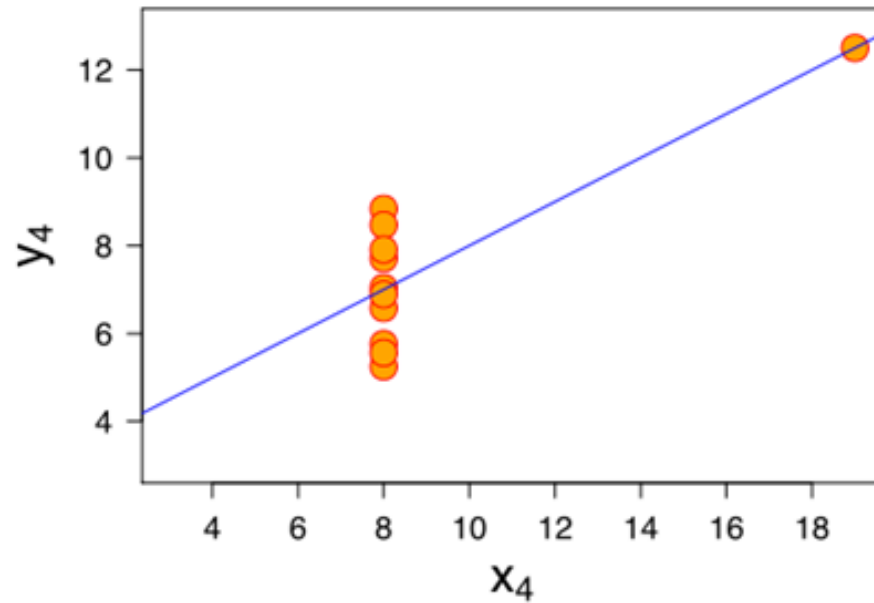
- i.e., do not extrapolate too far

Regression caution # 2

Plot the data! Regression lines are only appropriate when there is a linear trend in the data

Regression caution #3

Be aware of outliers – they can have an huge effect on the regression line



Linear regression in R

Regression lines in R

load the data

```
load("states_smoking.rda")
```

create a scatter plot and calculate the correlation

```
plot(smoking$CIG, smoking$LUNG)
```

Note: `plot(x, y)`



fit a regression model Note: `lm(y ~ x)`

```
lm_fit <- lm(smoking$LUNG ~ smoking$CIG)
```

get the a and b coefficients

```
coef(lm_fit)
```

add a regression line to the plot

```
abline(lm_fit)
```

Concepts for the relationship between two quantitative variables

A **scatterplot** graphs the relationship between two variables

The **correlation** is measure of the strength and direction of a linear association between two variables

- Value between -1 and 1

In **linear regression** we fit a line to the data, called the **regression line**

- We get coefficients for the slope (b) and the y-intercept (a)

The **residual** is the difference between an observed (y_i) and a predicted value ($\hat{y}_i$) of the response variable

- The regression line minimizes the sum of squared residuals

Sampling

Where do samples/data come from?



Example: sampling 100 sprinkles



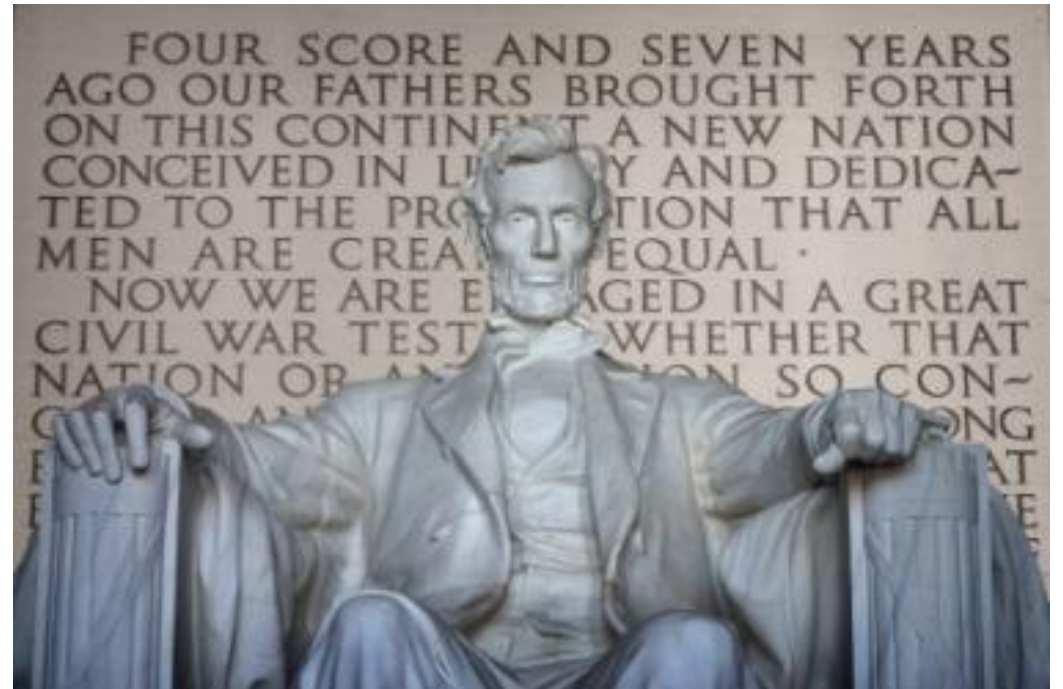
1	orange
2	red
3	green
4	white
5	white
6	white
7	white
8	white
9	red

The **sample size** (n) is the number of items in the sample
What is **n** in the sprinkle example?

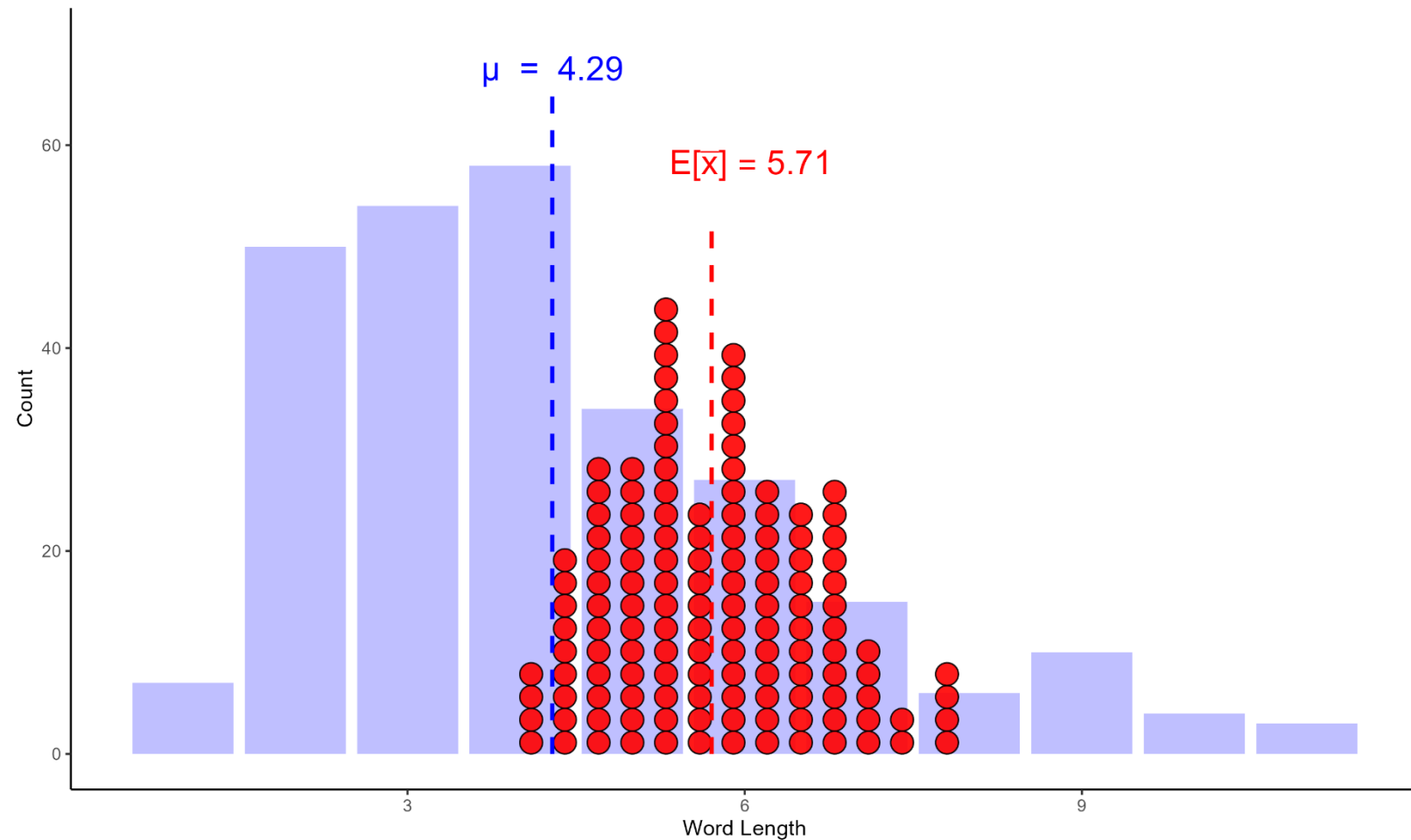
Sampling from the Gettysburg address

You filled out Gettysburg sampling worksheet survey on Canvas where you will randomly sample 10 words from the Gettysburg address

You reported the mean word $\bar{x}$ length of your 10 words



Bias and the Gettysburg address word length distribution



Bias

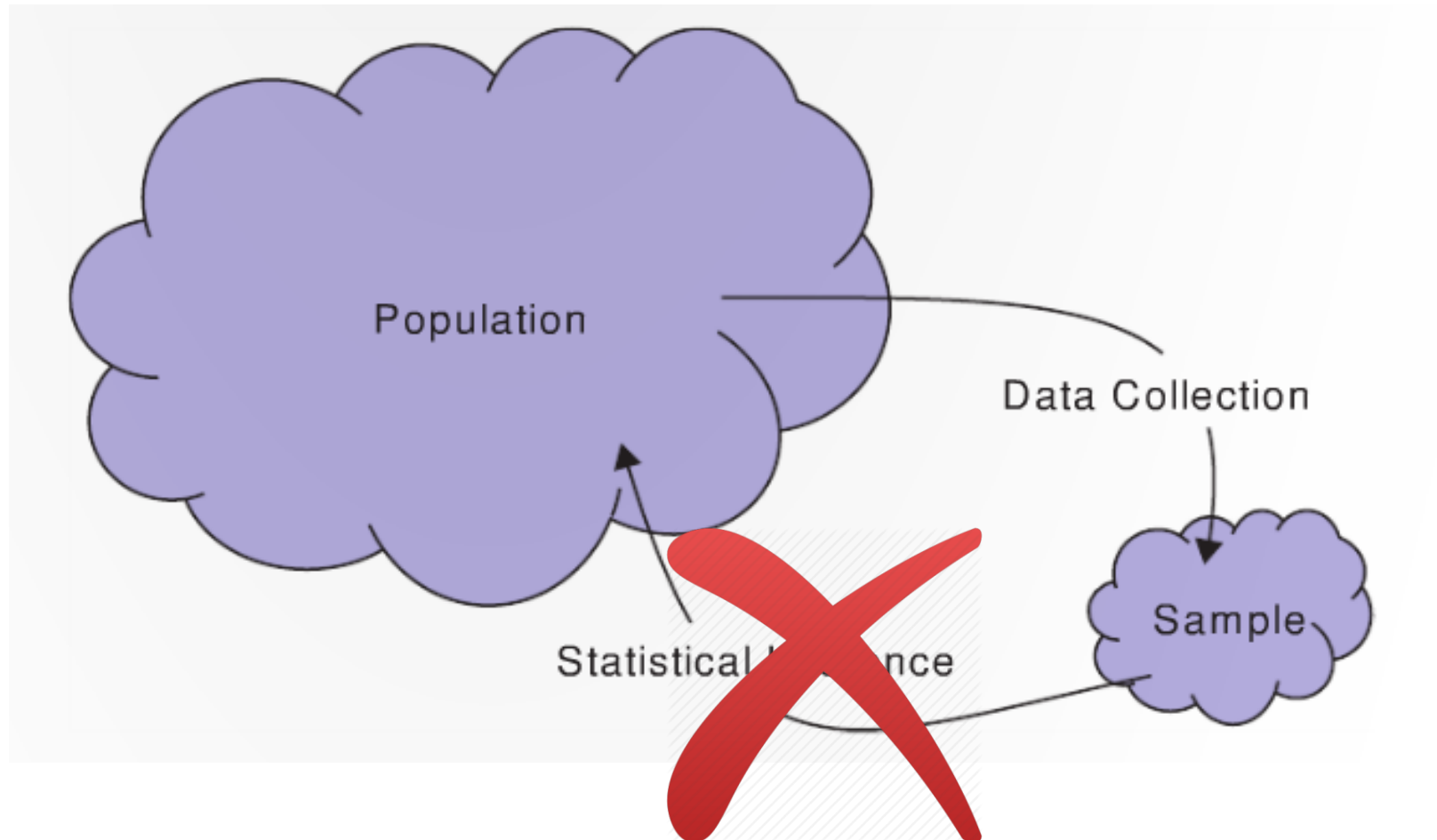
Sampling bias exists when the method of collecting the data causes the sample to inaccurately reflect the population

This leads to ***biased statistics*** where our average statistic value does not equal the parameter value

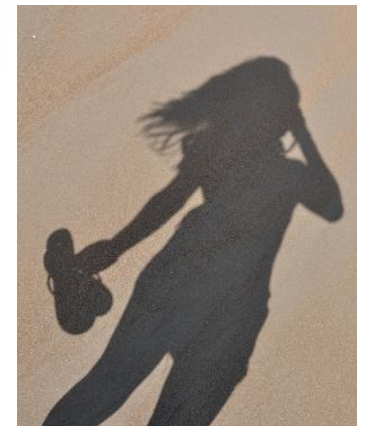
- E.g., $E_s[\bar{X}] \neq \mu$

Statistical bias

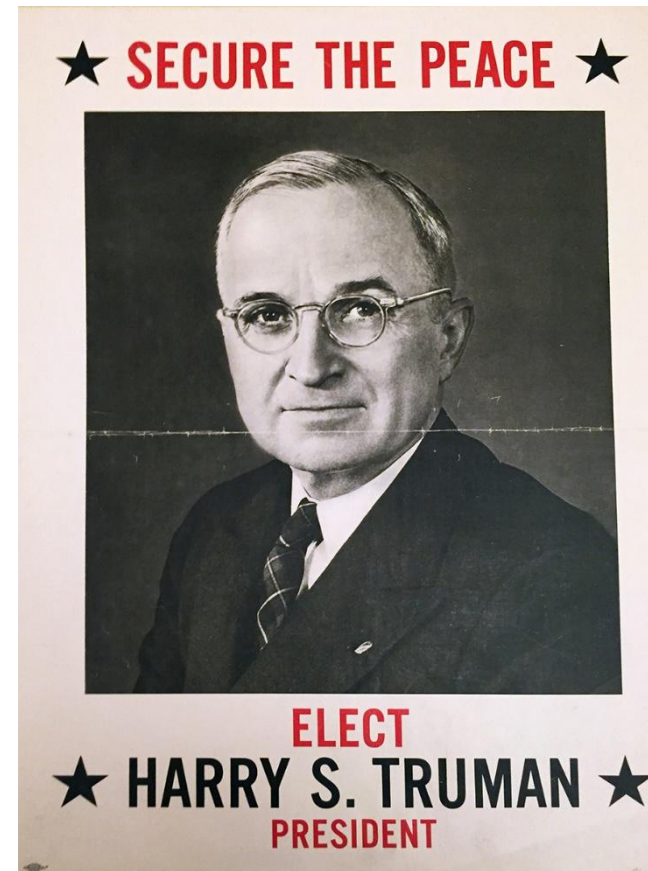
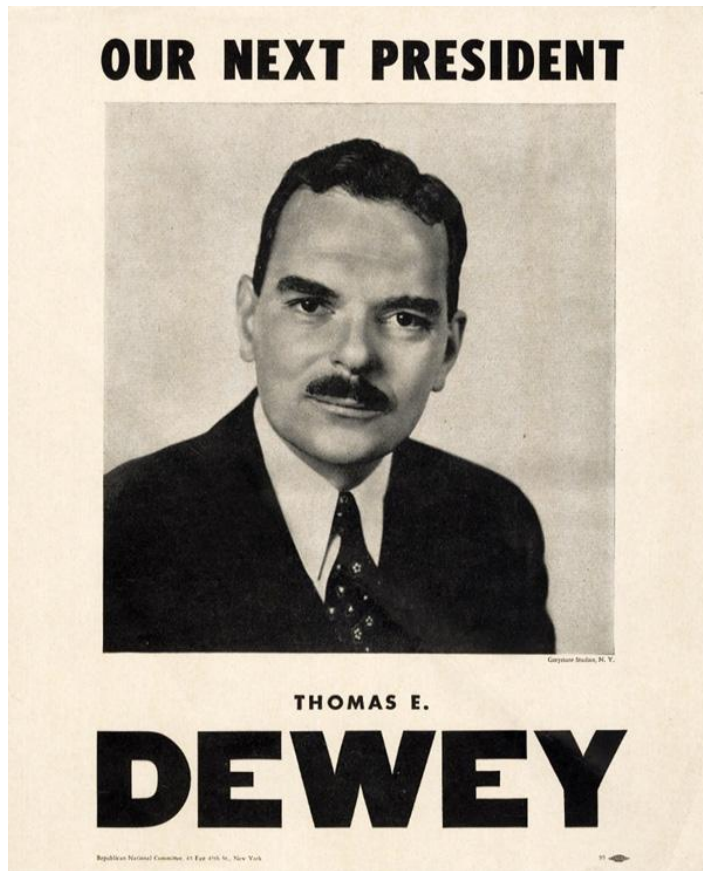
μ



$\bar{x}$



1948 US election



Newspaper title: Dewey Defeats Truman (1948)

The newspaper was published before the conclusion of the 1948 presidential election

The results were based on a large telephone poll which showed Dewey sweeping Truman

However, Harry S. Truman won the election

Q: What went wrong?



Basic questions for sampling

What is the population?

What is the sample?

Do they differ in a meaningful way?

To prevent bias: use simple random sample!

Simple random sample: each member in the population is equally likely to be in the sample

Allows for generalizations to the population!

Soup analogy



How do we select a random sample?

Mechanically:

- Flip coins

- Pull balls from well mixed bins

- Deal out shuffled cards, etc.

Use computer programs